Terafence
Reliable Transformations!

Bidirectional Security Gateway



TFG121-BSG



Technology Overview

Terafence is proud to offer a unique, secure, and innovative network security appliance – Terafence Bidirectional Gateway.

Terafence manufactures Cyber Security hardware products to allow security architects the freedom in protecting their networks as needed.

Terafence TFG121-BSG is designed to provide a better-then-ever security by filtering LAN traffic via Terafence's proprietary FPGA-based hardware architecture.

Terafence TFG121-BSG allows network architects to define traffic patterns through the device with total assurance that any hacking attempts fail due to galvanic separation and super strong appliance architecture.

Terafence TFG121-BSG utilizes dual-proprietary FPGA-based hardware architecture data diodes by Terafence to create an appliance that is protocol agnostic and yet extremely safe and secure.

Only pre-defined traffic can pass through without allowing direct TCP/IP sessions between client and server filtered by hardware-based traffic filters.

Why choose TFG121-BSG?

Key Features:

- Complete galvanic network isolation for enhanced security and protection.
- Acts as fully transparent proxy in the network
- Core security hardware operates independently without an operating system, MAC address, or IP address.
- Functions as Layer-2 Physical Firewall for inline traffic enforcement
- Near Real-time data transfer with minimal latency
- Secure, Electronically-controlled data exchange
- Operates within the same network segment only
- Hardware-based Packet filtering
- Port-based Management; Protocol Agnostic

Technical Specifications

Data Throughput:	Up to 1 Gbps with auto-negotiation
Power Supply:	100–240 VAC, 50/60 Hz, with dual redundant internal hot-swappable AC power supplies
Form Factor:	1U rack-mountable appliance
Operating Temperature:	32°F to 113°F
Storage Temperature:	-40 °F to 185 °F
Relative Humidity:	5% ~ 95% | 55°C, non-condensing
Processor:	Intel i3 or better
Dimensions:	(W × H × D): 500 × 440.2 × 44.2 mm (1U)
Cooling:	Dual fan cooling
Chassis:	IP30 protection, SECC, Black Metal
Reliability:	High availability, long-term operation, high MTBF (~10Years)
RAM –	Customizable as per requirement
Local Storage –	Customizable as per requirement
Network Interfaces:	1 x Data Port, 1 x Management Port per side
Relative Humidity:	- 5% ~ 95% | 131°F, non-condensing

In-door use only

Security Features

- Hardened Operating System on the integrated CPUs
- Secure Configuration & Management access (HTTPS) to GUI

All images are for illustrative purposes only. Product configurations, specifications, and features may be customized to meet customer-specific requirements and may vary accordingly.

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